# Tomoki Nishiyama

https://nishi24yama80.github.io/website/

## Education

**Graham School, the University of Chicago** — Chicago, USA
*Non-degree Program* — 2025 - Present
- Real Analysis and Environmental Economics

**Graduate School of Economics, The University of Tokyo** — Tokyo, Japan
*Master's Program in Economics* — 2024–2025 (on leave)
- A in first-year PhD Microeconomics, Macroeconomics, and Econometrics (A awarded to top 30%)
- Ranked 2nd out of 70 students in Econometrics I
- A in Measure Theory, Measure-Theoretic Mathematical Statistics, Measure-Theoretic Stochastic Processes, Graph Theory, and Computational Mathematics

**Department of Economics, The University of Tokyo** — Tokyo, Japan
*B.S. in Economics* — 2020 - 2024

## Awards and Honors

- **Brandt Research Fellow Award**, The best pre-doctoral research professional — 2026.09
  *Energy Policy Institute at UChicago*, *Award: $2,500*
- **Specially Selected Paper**, Thesis at the University of Tokyo — 2024.03
- **First Prize**, Empirical Micro Conference at Keio University — 2023.07

## Working Papers

**Inefficiencies of Revenue Sharing Auctions** — 2026.06 - Present

**Production Function Estimation with a Fixed Point Approach**
*Extension of Trunschke and Judd (2024)* — 2026.04 - Present

## Work in Progress

**Staying for Opportunity: Evidence from an In-Situ Housing Lottery in a Low-Income Community**
*with Andrew S. Griffen, Yukichi Mano, Yasuyuki Sawada, and Ze Wang*

**Screening in Public Procurement**
*with Ze Wang*

## Patent

**Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Water Usage**
*Japan Patent ID: JP-2024-018823* — 2024.02

**Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Utility Consumption Data (Water and Gas)**
*Japan Patent ID: JP-7137267* — 2022.09

## Job Experience

**Pre-doctoral Research Professional for Koichiro Ito** | Chicago, USA 2025.9 - Present
- Brandt Research Fellow (2026.9 - Now)
- Research Professional at Energy Policy Institute, the University of Chicago
- Becker Friedman Institute

**Research Assistant for Yasuyuki Sawada** | Tokyo, Japan 2022.07 - 2025.8
- Engaged in six field surveys in Manila, the Philippines.
- Prepared survey forms and manage the surveys with local collaborators in Manila.
- Constructed dataset from the field-survey results.
- Analyzed data with causal inference and deep-learning methods.

**Research Assistant at Hirofumi Uzawa International Institute** | Tokyo and Kyoto, Japan 2023.07 - 2025.8
- Constructed a model of environment economics and macroeconomics.
- Wrote academic articles for the general public.
- Made academic presentations at Kyoto University.

**Research Assistant for Taisuke Nakata** | Tokyo, Japan 2021.07 - 2021.12
- Constructed epidemic-macro models.
- Numerical simulations with Matlab.
- Prepared policy reports for Ministry of Health, Labour and Welfare.

**GIS Data Analyst at Microbase Inc.** | Tokyo Japan 2020.09 - 2023.07
- Data cleaning of administrative data and utility consumption data
- Geo-spatial analysis of the administrative data
- Development of machine-learning models of predicting vacant houses
- Optimization of future energy infrastructure distribution
- Prepare reports and presentations summarizing research findings
- Develop web-application

## Presentations

- Special Symposium "Evolution of Social Common Capital" (Kyoto University) 2024.03
- Japanese Association for Development Economics, Young Conference (Kyoto University) 2023.11
- Empirical Micro Conference (Keio University) 2023.07
- Korea-Japan Inter University Conference (Online) 2022.11

## Skills

**Languages**: Japanese (Native), English (Fluent), Mandarin Chinese (Intermediate).

**Programming**: Python, R, Julia, Matlab, LaTeX, HTML, SQL, GIS (GeoPandas, GDAL, QGIS, PostGIS), Git.